

# ANATOMY OF THE RETEST IN THE QQQ OPENING RANGE BREAKOUT

*Exploratory Evidence on Price Behavior Following the  
QQQ Opening Range Breakout*

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## ABSTRACT

This paper is an exploratory and descriptive study. The author makes no claims of causality or exploitability regarding the documented patterns.

We present an exploratory study of the post-breakout sequence in the Invesco QQQ Trust ETF, using intraday data at one-minute resolution from January 2017 through December 2024. The existing literature on the Opening Range Breakout focuses on entry at the moment of the breakout; the internal dynamics of the minutes that follow have not been measured to date.

We decompose the sequence into four levels — the opening range, the breakout, the retest, and the outcome — and document the statistical associations observed at each level. We find that range size is associated with the magnitude of the subsequent move, that a consistent fraction of breakouts generates a retest of the broken level within a defined time window, that the time elapsed until that retest shows the strongest association with the subsequent outcome, and that larger pre-retest excursions are associated with a lower probability of continuation — a counterintuitive pattern that we document along with its interpretive limitations.

All results are descriptive associations conditional on operational definitions that we document with precision. Causal mechanisms are unknown. We invite other researchers to replicate, challenge, and extend this work.

**Keywords:** *Opening Range Breakout, intraday momentum, breakout retest, QQQ, ETF microstructure*

# 1. INTRODUCTION

## 1.1 The Starting Point

The Opening Range Breakout is one of the most studied and longest-standing intraday trading strategies in financial markets. Its premise is simple: the price range established during the first minutes of the trading session reflects the accumulated order imbalance from the previous night, and a decisive move beyond that range signals sufficient directional conviction for the market to continue in that direction for the remainder of the day.

Despite its long history, the strategy retains both academic and practical relevance. The most recent and rigorous work — Zarattini and Aziz (2023) and Zarattini, Barbon, and Aziz (2024), distributed through the Swiss Finance Institute — documented that a five-minute ORB strategy applied to QQQ from 2016 through 2023 produced an annualized alpha of 33% net of commissions. These results place the ORB among the few market anomalies supported by solid academic evidence.

However, all prior literature shares one common feature: the position is initiated at the moment of the breakout and managed from that point forward. What occurs in the minutes following the breakout — whether price retraces to the level it just broke, whether that retrace has measurable properties, and whether those properties are associated with subsequent price behavior — has never been systematically measured. That is the gap this paper begins to address.

## 1.2 The Questions Guiding This Paper

This paper does not aim to demonstrate that the retest is exploitable in trading, nor that the patterns we find will persist in the future. The objective is more modest and, we believe, more honest: to measure for the first time what occurs after the ORB breakout in QQQ, with sufficient methodological rigor for other researchers to replicate, challenge, or extend the findings.

The four questions guiding the analysis are as follows. What fraction of ORB breakouts generates a retest of the broken level, under the conditions we define? What are the properties of the price excursion prior to that retest? How long does the retest take to occur? Are the conditions under which the retest occurs associated with subsequent price behavior, as classified by our definitions?

The explicit inclusion of the phrases "under the conditions we define" and "as classified by our definitions" in those questions is not accidental. It reflects the approach we adopt throughout this paper: being precise about what each number we report measures, and what it does not.

## 1.3 Why QQQ and Why This Period

QQQ was selected for three concrete reasons. First, it is one of the most liquid instruments in the global equity market, with bid-ask spreads at one-minute resolution typically ranging from \$0.01 to \$0.02, allowing price dynamics to be studied without the microstructure noise that contaminates less liquid assets. Second, it tracks the Nasdaq-100, with massive participation from both institutional and individual investors, ensuring that the observed patterns reflect genuine market behavior with sufficient depth. Third, the most relevant prior literature uses QQQ as its primary instrument, enabling direct comparisons with existing work.

The 2017–2024 period spans five qualitatively distinct market regimes: a low-volatility bull market (2017–2019), the COVID-19 crisis and subsequent recovery (2020), an accelerated bull market (2021), a sustained bear market driven by the Federal Reserve's monetary tightening with a 32.6%

decline (2022), and a recovery to new highs (2023–2024). This diversity of conditions is important for evaluating whether the observed patterns are specific to a particular type of market or appear with broader generality.

## **1.4 Scope of the Study**

This paper is a descriptive and exploratory study of a single instrument over a specific period. It neither implements nor evaluates trading strategies. It does not incorporate transaction costs or execution slippage. It makes no claim that the patterns found are causal or that they will persist outside the sample studied.

A caveat the reader should keep in mind from the outset: in a study such as this, operational definitions are not neutral conventions. They are decisions that determine what is measured and, therefore, what is found. In particular, the way we define the Continuation outcome and the Failure outcome can mechanically influence some of the numerical results we report. We document these effects with precision in Section 3 and revisit them each time they are relevant. This transparency about the limitations of our own definitions is, in our view, an essential part of the contribution this paper seeks to make.

## **1.5 Structure of the Paper**

Section 2 reviews the related literature. Section 3 describes the data, the analytical framework, and, with particular care, the operational definitions of all measured events and their possible effects on the results. Sections 4 through 7 present the results for each level of the analysis. Section 8 presents the multivariate analysis. Section 9 presents the robustness analysis. Section 10 discusses the observed patterns, their possible interpretations as unverified hypotheses, and their limitations. Section 11 concludes with the questions this paper leaves open for future research.

# **2. RELATED LITERATURE**

## **2.1 The ORB in Academia**

Academic research on the ORB begins formally with Holmberg, Lönnbark, and Lundström (2012), who proposed a statistical framework for evaluating the profitability of ORB strategies using daily OHLC data. Applied to crude oil futures from 1983 to 2011, they found significantly positive average returns concentrated in periods of elevated volatility. Lundström (2017) extended this finding, documenting that ORB profitability grows monotonically with realized volatility. This result is relevant as context for the patterns we observe in Level 1 regarding opening range size.

Tsai, Wang, and collaborators (2018) introduced the concept of the Timely Opening Range Breakout, using one-minute data across five index futures markets from 2003 to 2013. Their main finding was that the optimal range formation period is short — typically five minutes or less — and that the quality of the breakout signal decays rapidly with the time of day. This is consistent with the patterns we observe in Level 2.

The most directly relevant work is the series of papers by Zarattini and collaborators. Zarattini and Aziz (2023) investigated ORB profitability in QQQ using a five-minute range, documenting an annualized alpha of 33% and a Sharpe ratio above 2.0. Zarattini, Barbon, and Aziz (2024) extended the analysis to more than 7,000 U.S. stocks.

None of these papers studies the post-breakout retest. Entry occurs on the breakout candle and position management begins from that point. The internal dynamics of the minutes that follow are not addressed in any prior work known to the authors.

## **2.2 Intraday Microstructure as Theoretical Context**

Chordia, Roll, and Subrahmanyam (2002) documented that order imbalances at the start of the session are substantially larger than at other times of day. This is consistent with the concentration of breakouts at the open that we document in Level 2, although it does not establish causality between the two phenomena.

Dim, Eraker, and Vilkov (2023) documented that market maker inventory gamma predicts intraday price reversals. We mention this work as context for the interpretive hypotheses proposed in the Discussion. We do not verify those hypotheses directly, and it should not be understood as evidence that the mechanism operates in our data.

## **2.3 Positioning of This Paper**

This paper differs from the existing literature in three concrete respects. First, it disaggregates the post-breakout sequence into measurable components with statistical properties that can be studied separately. Second, the retest is operationalized using a precise, pre-specified definition that allows for reproducibility. Third, the questions are descriptive rather than prescriptive — we seek not the optimal strategy but the properties of the phenomenon as it appears in the data under our definitions.

# **3. DATA AND METHODOLOGY**

## **3.1 The Thinking Behind the Design**

Before formally describing the data, it is worth explaining the logic that led to this design. The original question was simple: after price breaks the QQQ opening range, what happens?

The most immediate answer is that price sometimes continues in the direction of the breakout and sometimes reverses. But that is too vague to serve as a useful scientific starting point. What we want to know is whether there is something observable — in the range that preceded the breakout, in the breakout itself, or in the minutes that follow — that is associated with one outcome or the other.

To answer that in an orderly way, we designed a framework that moves from the general to the specific. It makes no sense to jump directly to the outcome without first characterizing the context in which that outcome occurs. That chain of reasoning led to the design of four sequential levels: first the range, then the breakout, then the retest, then the outcome. Each level builds on the previous one, and its results serve as input for the next.

## **3.2 The Data**

The dataset consists of one-minute OHLCV bars — open, high, low, close, and volume — for the Invesco QQQ Trust from January 26, 2017 through December 31, 2024, covering the regular trading session from 9:30 AM to 3:59 PM Eastern Time. All timestamps are converted to Eastern Time with explicit UTC offset handling to ensure correct treatment of daylight saving time transitions.

### 3.3 Why One-Minute Data

The choice of one-minute resolution reflects a balance between granularity and noise. Daily data does not allow the internal sequence of the breakout to be studied. Tick or second-level data adds microstructure noise that complicates the identification of meaningful events without providing substantive additional information for the questions of this paper. One-minute resolution allows precise measurement of when the breakout occurs, when the retest occurs, and how long each event takes, while filtering out smaller-scale fluctuations.

### 3.4 The Four-Level Analytical Framework

The analysis is organized into four sequential levels, where each level builds on the outputs of the previous one. This architecture reflects the natural temporal sequence of the phenomenon: first the range forms, then the breakout occurs, then the retest may occur, and finally the subsequent price behavior is observed. The goal of this structure is to identify at which point in the sequence the information associated with the outcome is concentrated, and whether that information is statistically independent across levels.

### 3.5 Opening Range Durations

All analyses are conducted in parallel for 24 durations from 5 to 120 minutes in 5-minute increments. This decision reflects the absence of prior empirical consensus on the optimal range duration for QQQ specifically. Results are reported for the 15-minute range as the representative case, selected because it yields an adequate sample size and a formation period long enough to absorb the most volatile minutes of the open. Where patterns differ materially across durations, this is documented explicitly.

### 3.6 Operational Definitions and Their Relationship to the Results

This is the most important section of the paper for any reader who wishes to interpret the results correctly. In an empirical study such as this, operational definitions are not merely technical conventions: they are decisions that determine what enters the sample, how each observation is classified, and consequently what numbers appear in the tables. A result that appears to describe a market phenomenon may in part reflect the properties of the definition used to measure it.

We present each definition alongside an explicit note on how it may affect the results. All definitions were fixed prior to running any analysis, which is important for transparency, but does not eliminate the effects described below.

**Opening Range (OR):** The price range defined by the highest high and lowest low of all one-minute bars from 9:30 AM to 9:30 AM + N minutes, where  $N \in \{5, 10, 15, \dots, 120\}$ .

**Breakout:** The first one-minute bar whose high exceeds the OR high (bullish breakout) or whose low falls below the OR low (bearish breakout). Only the first breakout of the day is analyzed for each range duration.

*Note on the selection bias introduced by this definition:* By analyzing only the first breakout of the day, we exclude days with multiple breakouts and days with no breakout. The resulting sample does not represent all QQQ trading days but a subset with specific properties. The patterns we document apply to that subset, and their generalization to other contexts requires caution.

**Valid Retest:** A post-breakout event that satisfies three conditions sequentially and in order.

*Condition 1 — Minimum 0.20% excursion:* After the breakout, price must reach a minimum favorable excursion of 0.20% of the breakout price before any pullback is considered a valid retest. This threshold was chosen to separate genuine retests from bars that simply oscillate around the level without price having established a defined direction. At QQQ prices between \$130 and \$500, 0.20% represents between \$0.26 and \$1.00.

*Condition 2 — Level contact,  $\pm 0.05\%$  tolerance:* After satisfying Condition 1, price must return to the breakout level within a tolerance of 0.05% of the breakout price. This tolerance acknowledges that price levels have finite width in practice and an exact price is rarely touched to the cent.

*Condition 3 — 120-minute window:* The retest must occur within 120 minutes of the breakout. Retests occurring after that window are considered to belong to a different intraday context.

*Note on the sensitivity of these conditions:* The robustness analysis in Section 9 shows that the relative gap between fast and slow retests ranges from 36.7 to 48.3 percentage points across all threshold specifications explored. This suggests that the temporal pattern of the retest does not depend critically on the exact values chosen, although the absolute figures change with the thresholds.

**Continuation Outcome:** Within 60 minutes after the retest, price exceeds the maximum favorable pre-retest excursion — that is, it surpasses the furthest point it had reached between the breakout and the retest. We refer to that point as MFE\_pre.

*Note on the mechanical effect of this definition:* This is the most important definition to understand in terms of its implications for the results. The bar that price must clear to be classified as a Continuation is the MFE\_pre. When the MFE\_pre is small — for example, 0.25% — that bar is relatively easy to clear. When the MFE\_pre is large — for example, 0.80% — the bar is substantially higher and harder to reach within the 60-minute window. This creates a mechanical relationship between MFE\_pre and the Continuation rate: the larger the MFE\_pre, the more demanding the definition, regardless of what the market does.

The fourth pattern we document in Level 4 — that larger MFE\_pre is associated with a lower Continuation rate — may be partially or entirely produced by this mechanical effect. We cannot determine with certainty how much of that pattern reflects a genuine market phenomenon and how

much reflects the increasing difficulty imposed by the definition. We report the pattern because it is statistically robust and because the question of what produces it is itself valuable, but the reader should keep this caveat in mind when interpreting that result.

One way to disentangle the two effects in future research would be to redefine Continuation using a fixed target — for example, an additional 0.20% move beyond the retest price — rather than the variable target used here. If the MFE\_pre pattern persists under that alternative definition, it would constitute stronger evidence that it reflects a genuine market phenomenon.

**Failure Outcome:** Within 60 minutes after the retest, price crosses the opposite boundary of the opening range — OR\_low for bullish breakouts, OR\_high for bearish breakouts — before achieving a Continuation Outcome.

*Note on the mechanical effect of this definition:* The failure barrier is the opposite extreme of the OR. When the opening range is wider, that opposite extreme is farther from the breakout price. Therefore, the Failure definition requires a larger move on days with wide ranges than on days with narrow ranges. This may explain, at least in part, the observed pattern whereby larger ranges are associated with lower Failure rates — not necessarily because the phenomenon is different, but because the definition places the barrier farther away. The reader should keep this in mind when interpreting the results in Table 10 and Table 13.

**Neutral Outcome:** Neither of the above occurs within 60 minutes of the retest. The analysis in Section 9 shows that these cases resolve approximately 50/50 between Continuation and Failure when the window is extended to the session close, suggesting they represent genuine market indecision within the original window.

### 3.7 Sample Flow Diagram

To facilitate exact replication of the analysis, we document the filtering process step by step for the 15-minute OR case, which is the representative case used throughout the paper.

The dataset begins with 2,290 trading days. Of these, 2,288 days yield a valid opening range with sufficient bars for full formation. Of those 2,288 days, all produce at least one OR breakout during the session, but only the first is analyzed. Of the 2,288 breakouts, 1,565 (68.4%) meet the minimum 0.20% excursion criterion in the direction of the breakout before retracing. Of those 1,565, a total of 1,045 (66.8%) generate a valid retest within the 120-minute window. Those 1,045 retests constitute the final Level 4 sample. Of them, 482 (46.1%) produce a Continuation Outcome, 195 (18.7%) produce a Failure Outcome, and 368 (35.2%) produce a Neutral Outcome within the 60-minute window.

Differences in N across tables in the paper — 2,290, 2,288, 2,277, and other values — reflect this sequential filtering process across different analyses and OR durations. Each table reports the N corresponding to the specific subset analyzed in that table.

### 3.8 Variable Table

*Table 1 — Complete Variable Definitions*

| Variable                  | Level | Definition                                                                   | Unit        |
|---------------------------|-------|------------------------------------------------------------------------------|-------------|
| <b>OR_size_pct</b>        | 1     | $(OR\_high - OR\_low) / OR\_low \times 100$                                  | %           |
| <b>OR_close_position</b>  | 1     | $(close\_last\_OR\_bar - OR\_low) / OR\_size$                                | 0–1         |
| <b>gap_pct</b>            | 1     | $(open\_day - close\_previous\_day) / close\_previous\_day \times 100$       | %           |
| <b>breakout_minute</b>    | 2     | Minutes from session open to the breakout bar                                | min         |
| <b>vol_ratio</b>          | 2     | $breakout\_bar\_volume / mean(OR\_bars\_volume)$                             | ratio       |
| <b>mfe_pre_retest_pct</b> | 3     | Maximum excursion between breakout and retest / $breakout\_price \times 100$ | %           |
| <b>mins_to_retest</b>     | 3     | Minutes from the breakout bar to the retest bar                              | min         |
| <b>outcome</b>            | 4     | Continuation / Failure / Neutral per definitions in Section 3.6              | categorical |

## 4. LEVEL 1 — ANATOMY OF THE OPENING RANGE

### 4.1 The Question and Why We Start Here

The first level aims to characterize the opening range before any breakout occurs. The reason for starting here is that the range is the context in which everything else takes place. If we want to understand what is associated with price behavior after the retest, we first need to understand the conditions under which that retest occurs. The opening range is the initial state of the system.

### 4.2 Pattern 1.1 — Range Size Is Linearly Associated with Maximum Post-Breakout Excursion

**Table 2 — OR Size Quintiles and MFE by Quintile (OR = 15 minutes, N = 2,128 days with breakout)**

| Quintile             | OR_size% | N   | %Up   | %Down | MFE_avg% | MFE_med% | ClosePos_Up | ClosePos_Down |
|----------------------|----------|-----|-------|-------|----------|----------|-------------|---------------|
| <b>Q1</b><br>minimum | 0.23%    | 426 | 50.7% | 49.3% | 0.41%    | 0.30%    | 0.658       | 0.402         |
| <b>Q2 small</b>      | 0.33%    | 426 | 54.0% | 46.0% | 0.53%    | 0.43%    | 0.703       | 0.335         |
| <b>Q3</b><br>medium  | 0.43%    | 426 | 53.8% | 46.2% | 0.59%    | 0.43%    | 0.716       | 0.340         |
| <b>Q4 large</b>      | 0.57%    | 426 | 51.9% | 48.1% | 0.74%    | 0.57%    | 0.722       | 0.302         |
| <b>Q5 very large</b> | 0.87%    | 426 | 49.5% | 50.5% | 1.15%    | 0.89%    | 0.733       | 0.248         |

In the observed data, the average MFE — the maximum price excursion from the breakout level to the session close — grows monotonically from 0.41% in the minimum range quintile to 1.15% in the largest, a difference of 2.8 times. This association holds without exception across every quintile and across all 24 OR durations studied ( $Z=+7.109$ ,  $p<0.001$ ).

What this association describes, in concrete terms, is that days on which QQQ establishes a wider opening range tend to produce breakouts with greater subsequent excursion, under the MFE definition we use. Why this association occurs is a separate question that these data do not answer. One possible hypothesis is that range size reflects the degree of imbalance between buyers and sellers during range formation, and that this imbalance persists after the breakout. But this is an interpretive hypothesis, not a result.

#### **4.3 Pattern 1.2 — The Closing Position of the OR Within the Range Is Associated with the Direction of the Subsequent Breakout**

The last two columns of Table 2 show where the OR formation period closed within its own range, separated by the direction of the breakout that subsequently occurred. The OR\_close\_position variable takes a value of 0 when the close is exactly the range low and 1 when it is exactly the range high.

What is observed is consistent across all quintiles: when the subsequent breakout was bullish, the OR had closed on average in the upper 65–73% of the range. When the breakout was bearish, it had closed in the lower 25–40%. This information is observable before the breakout. However, the distribution of bullish versus bearish breakouts is approximately 50/50 across all quintiles, indicating that this variable does not predict direction with sufficient certainty to constitute a directional signal on its own.

#### 4.4 Pattern 1.3 — Days with a Bearish Gap Are Associated with a Higher Average MFE

*Table 3 — Effect of the Overnight Gap on MFE (OR = 15 minutes)*

| Gap Type                       | N     | %Break Up | %Break Down | MFE_avg% | MFE_med% |
|--------------------------------|-------|-----------|-------------|----------|----------|
| <b>Bullish gap (&gt;+0.1%)</b> | 1,040 | 50.7%     | 49.3%       | 0.65%    | 0.48%    |
| <b>Bearish gap (&lt;-0.1%)</b> | 775   | 54.1%     | 45.9%       | 0.79%    | 0.54%    |
| <b>Flat gap (±0.1%)</b>        | 314   | 51.3%     | 48.7%       | 0.55%    | 0.39%    |

In the data, days that open with a bearish gap show an average MFE of 0.79%, higher than the 0.65% observed on days with a bullish gap. One possible hypothesis is that bearish gaps concentrate greater uncertainty and volatility that persists through the session. We did not verify this hypothesis directly.

#### 4.5 Pattern 1.4 — Breakout Direction Is Approximately Equiprobable

Across all range size quintiles, the fraction of bullish versus bearish breakouts ranges from 49.5% to 54.0%. Range size is not associated with breakout direction — only with the magnitude of the subsequent excursion. This observation establishes that the ORB as a phenomenon does not resolve the problem of direction.

## 5. LEVEL 2 — ANATOMY OF THE BREAKOUT

### 5.1 The Question

Once we understand the range, the natural next question is whether there is something in the breakout itself that is associated with the subsequent move. This distinction matters because in practice an observer does not only see the range that has formed — they also see how the breakout occurs: at what time, and with what relative volume.

## 5.2 Pattern 2.1 — 82% of Breakouts Are Concentrated in the First 30 Minutes of the Session

*Table 4 — MFE by Breakout Time (OR = 15 minutes)*

| Time Window   | N     | %Up   | MFE_avg% | MFE_med% | Vol_ratio_avg |
|---------------|-------|-------|----------|----------|---------------|
| 09:30 – 10:00 | 1,870 | 51.8% | 0.71%    | 0.50%    | 0.95          |
| 10:00 – 10:30 | 351   | 53.8% | 0.67%    | 0.43%    | 0.90          |
| 10:30 – 11:00 | 44    | 63.6% | 0.48%    | 0.36%    | 0.76          |
| 11:00 – 11:30 | 12    | 41.7% | 0.48%    | 0.30%    | 0.76          |

Of the 2,277 breakouts analyzed for the 15-minute OR, 1,870 (82.1%) occur within the 09:30–10:00 window. Average MFE decreases with time of day: 0.71% in the first window versus 0.48% after 10:30. Late breakouts are not only less frequent but are also associated with shorter average excursions. This is consistent with the findings of Chordia et al. (2002) and Tsai et al. (2018) on the concentration of price discovery at the open, although our measurement is descriptive and does not establish the same causal relationship as those studies.

## 5.3 Pattern 2.2 — Vol\_ratio Shows a Non-Linear Association with MFE

*Table 5 — MFE by Vol\_ratio Quintile (OR = 15 minutes)*

| Quintile     | Vol_ratio_avg | N   | MFE_avg% | MFE_med% |
|--------------|---------------|-----|----------|----------|
| Q1 — lowest  | 0.71          | 458 | 0.53%    | 0.36%    |
| Q2           | 0.88          | 457 | 0.68%    | 0.50%    |
| Q3 — optimal | 0.95          | 458 | 0.80%    | 0.57%    |
| Q4           | 1.01          | 457 | 0.75%    | 0.52%    |
| Q5 — highest | 1.15          | 458 | 0.71%    | 0.49%    |

The association is statistically significant ( $\chi^2=26.85$ ,  $p=0.00075$ ) but non-linear in form: average MFE peaks at quintile Q3 and declines at Q5. Breakouts with the highest relative volume show lower average MFE than those with moderate-to-high volume. One possible hypothesis is that very high volume may indicate that the move is being absorbed by participants on the opposite side, but this

interpretation was not verified directly. What the data show is the association; the mechanism is a separate question.

#### 5.4 Pattern 2.3 — Day of the Week Shows No Statistically Significant Association

*Table 6 — MFE by Day of the Week (OR = 15 minutes)*

| Day       | N   | %Up   | MFE_avg% | MFE_med% |
|-----------|-----|-------|----------|----------|
| Monday    | 428 | 58.9% | 0.66%    | 0.44%    |
| Tuesday   | 471 | 51.2% | 0.68%    | 0.50%    |
| Wednesday | 467 | 51.0% | 0.69%    | 0.49%    |
| Thursday  | 460 | 48.3% | 0.73%    | 0.49%    |
| Friday    | 462 | 52.8% | 0.71%    | 0.50%    |

The chi-square test yields  $\chi^2=4.63$ ,  $p=0.796$ . The descriptive differences across days are statistically indistinguishable from chance. We include this analysis because scientific honesty requires reporting what was sought and not found.

## 6. LEVEL 3 — ANATOMY OF THE RETEST

### 6.1 Previously Unmeasured Territory

Levels 1 and 2 characterized the context of the breakout. Level 3 is where this paper enters genuinely unmeasured territory in the prior literature. We found no study that has quantified what fraction of ORB breakouts generates a retest under precisely defined conditions, how long that retest takes to occur, or how the price excursion before it is distributed.

### 6.2 Pattern 3.1 — Observed Retest Rates by OR Duration

*Table 7 — Retest Statistics by OR Duration (selected durations)*

| OR_<br>min | N_break<br>outs | N_excur<br>sion | N_ret<br>est | %Excur<br>sion | %Retest/Exc<br>ursion | MFE_pre_<br>avg% | Mins_retes<br>t_avg |
|------------|-----------------|-----------------|--------------|----------------|-----------------------|------------------|---------------------|
| 5          | 2,290           | 1,583           | 1,096        | 69.1%          | 69.2%                 | 0.361%           | 33.5                |
| 10         | 2,289           | 1,538           | 1,037        | 67.2%          | 67.4%                 | 0.349%           | 33.3                |

|            |       |       |       |       |       |        |      |
|------------|-------|-------|-------|-------|-------|--------|------|
| <b>15</b>  | 2,288 | 1,565 | 1,045 | 68.4% | 66.8% | 0.351% | 35.8 |
| <b>30</b>  | 2,274 | 1,484 | 966   | 65.3% | 65.1% | 0.351% | 38.4 |
| <b>60</b>  | 2,198 | 1,316 | 810   | 59.9% | 61.6% | 0.336% | 41.0 |
| <b>120</b> | 1,953 | 1,057 | 638   | 54.1% | 60.4% | 0.348% | 48.7 |

For the 15-minute OR: of 2,288 breakouts, 1,565 (68.4%) reached the defined minimum excursion of 0.20%. Of those, 1,045 (66.8%) generated a valid retest under the three defined conditions within 120 minutes. The rate decreases modestly with OR duration. These percentages are descriptive of what occurs in the data under the conditions we define; they are not rates of anything that would occur under different definitions.

### 6.3 Pattern 3.2 — MFE\_pre Is Monotonically Associated with the Probability of Retest

**Table 8 — Retest Rate by MFE\_pre Bin (OR = 15 minutes)**

| <b>MFE_pre Bin</b>  | <b>N</b> | <b>N_retest</b> | <b>%Retest</b> | <b>Avg_mins</b> |
|---------------------|----------|-----------------|----------------|-----------------|
| <b>0.20 – 0.30%</b> | 1,041    | 560             | 53.8%          | 28.3            |
| <b>0.30 – 0.40%</b> | 274      | 242             | 88.3%          | 36.9            |
| <b>0.40 – 0.50%</b> | 108      | 103             | 95.4%          | 44.6            |
| <b>0.50 – 0.75%</b> | 94       | 93              | 98.9%          | 56.4            |
| <b>0.75 – 1.00%</b> | 32       | 31              | 96.9%          | 58.9            |
| <b>1.00 – 1.50%</b> | 14       | 14              | 100.0%         | 58.3            |

In the data, the greater the price excursion prior to the retest, the higher the probability that the retest occurs within the defined window. Beyond a 0.50% excursion, the retest is virtually certain. One interpretive hypothesis is that larger moves generate imbalances that the market tends to correct. This hypothesis was not verified directly.

### 6.4 Pattern 3.3 — Faster Retests Are Associated with Smaller MFE\_pre

**Table 9 — Temporal Distribution of the Retest (OR = 15 minutes, N = 1,045 retests)**

| <b>Minutes to Retest</b> | <b>N</b> | <b>% of Total</b> | <b>MFE_pre_avg%</b> |
|--------------------------|----------|-------------------|---------------------|
|--------------------------|----------|-------------------|---------------------|

|                     |     |       |        |
|---------------------|-----|-------|--------|
| <b>0 – 5 min</b>    | 75  | 7.2%  | 0.257% |
| <b>5 – 10 min</b>   | 138 | 13.2% | 0.284% |
| <b>10 – 15 min</b>  | 119 | 11.4% | 0.305% |
| <b>15 – 20 min</b>  | 107 | 10.2% | 0.311% |
| <b>20 – 30 min</b>  | 136 | 13.0% | 0.338% |
| <b>30 – 45 min</b>  | 143 | 13.7% | 0.372% |
| <b>45 – 60 min</b>  | 100 | 9.6%  | 0.420% |
| <b>60 – 90 min</b>  | 136 | 13.0% | 0.406% |
| <b>90 – 120 min</b> | 87  | 8.3%  | 0.461% |

The temporal distribution is relatively flat between 5 and 90 minutes. The observable pattern is that retests occurring more quickly have smaller MFE\_pre, while later retests have larger MFE\_pre. This pattern is relevant for interpreting the Level 4 results, but it also carries an important methodological implication: since MFE\_pre mechanically influences the Continuation definition, fast retests face a lower continuation bar than late ones. The reader should keep this in mind when reading Table 11.

## 7. LEVEL 4 — THE OUTCOME

### 7.1 The Central Question and Its Accompanying Caveat

We now reach the level where the central question of the paper is answered: given that a valid retest occurred under our definitions, what is observed in the following 60 minutes, as classified by our outcome definitions?

This framing of the question — explicitly conditional on the definitions — is intentional. The numbers we present at this level are descriptive associations under specific conditions, not absolute properties of the market. Any reader wishing to compare these results with other studies or apply them to other contexts must first verify that the definitions are comparable.

## 7.2 Base Outcome Distribution

**Table 10 — Outcome Distribution by OR Duration (selected durations)**

| OR_min | N_retests | %Cont | %Failure | %Neutral | MFE_post_<br>avg% | MAE_failure_<br>avg% |
|--------|-----------|-------|----------|----------|-------------------|----------------------|
| 5      | 1,096     | 47.7% | 32.7%    | 19.6%    | 0.368%            | 0.412%               |
| 10     | 1,037     | 47.8% | 24.3%    | 27.9%    | 0.364%            | 0.522%               |
| 15     | 1,045     | 46.1% | 18.7%    | 35.2%    | 0.361%            | 0.584%               |
| 30     | 966       | 47.9% | 8.6%     | 43.5%    | 0.357%            | 0.641%               |
| 60     | 810       | 44.2% | 2.1%     | 53.7%    | 0.346%            | 0.889%               |
| 120    | 638       | 42.2% | 1.1%     | 56.7%    | 0.363%            | 1.137%               |

For the 15-minute OR: 482 (46.1%) of the 1,045 retests produce a Continuation Outcome, 195 (18.7%) produce a Failure Outcome, and 368 (35.2%) produce a Neutral Outcome within the 60-minute window.

The most striking pattern in this table is the dramatic decline in the Failure rate as OR duration increases — from 32.7% at OR=5 min to 1.1% at OR=120 min. As documented in Section 3.6, part of this decline is a consequence of the definition: longer ranges place the Failure barrier farther away, making it mechanically harder to trigger. We cannot precisely quantify what fraction of this decline is a definitional effect and what fraction reflects a genuine market phenomenon.

## 7.3 Pattern 4.1 — Time to Retest Is Negatively Associated with the Continuation Rate

**Table 11 — Continuation Rate by Time to Retest (OR = 15 minutes)**

| Minutes to Retest | N   | %Continuation | 95% Wilson CI  | %Failure | %Neutral |
|-------------------|-----|---------------|----------------|----------|----------|
| 0 – 10 min        | 213 | 69.0%         | [62.5%, 74.8%] | 20.2%    | 11.8%    |
| 10 – 20 min       | 226 | 54.0%         | [47.5%, 60.4%] | 23.9%    | 23.9%    |
| 20 – 30 min       | 136 | 47.1%         | [38.9%, 55.4%] | 21.5%    | 33.1%    |
| 30 – 45 min       | 143 | 39.2%         | [31.5%, 47.3%] | 15.3%    | 44.4%    |
| 45 – 60 min       | 100 | 35.0%         | [26.4%, 44.8%] | 24.5%    | 42.6%    |

|          |     |       |                |       |       |
|----------|-----|-------|----------------|-------|-------|
| > 60 min | 227 | 25.6% | [20.3%, 31.6%] | 10.0% | 63.8% |
|----------|-----|-------|----------------|-------|-------|

$\chi^2=170.67$ ,  $p<0.001$  | Cochran-Armitage  $Z=-9.708$ ,  $p<0.001$

In the data, the rate of outcomes classified as Continuation decreases monotonically with time to retest, without exception across any bin. The confidence intervals of the extremes do not overlap: the upper bound of the CI for late retests (31.6%) is below the lower bound for fast retests (62.5%). This is the most statistically robust pattern in the paper.

This pattern is not subject to the mechanical effect of the MFE\_pre definition documented in Section 3.6, as time to retest and the Continuation bar are independent variables. Section 8 further confirms that this pattern is independent of breakout time in the logistic regression.

#### 7.4 Pattern 4.2 — MFE\_pre Is Negatively Associated with the Continuation Rate

**Table 12 — Continuation Rate by Pre-Retest Excursion (OR = 15 minutes)**

| MFE_pre Bin  | N   | %Continuation | 95% Wilson CI  | %Failure | %Neutral |
|--------------|-----|---------------|----------------|----------|----------|
| 0.20 – 0.30% | 560 | 54.6%         | [50.5%, 58.7%] | 17.7%    | 27.5%    |
| 0.30 – 0.40% | 242 | 46.3%         | [40.1%, 52.6%] | 18.6%    | 35.1%    |
| 0.40 – 0.50% | 103 | 34.0%         | [25.6%, 43.6%] | 19.4%    | 46.6%    |
| 0.50 – 0.75% | 93  | 22.6%         | [15.3%, 32.1%] | 20.4%    | 57.0%    |
| 0.75 – 1.00% | 31  | 22.6%         | [11.4%, 39.8%] | 20.4%    | 57.0%    |
| > 1.00%      | 16  | 6.2%*         | [1.1%, 28.3%]* | 25.5%    | 57.4%    |

\*N=16, low statistical power. Report with caution. Cochran-Armitage  $Z=-7.697$ ,  $p<0.001$

In the data, the rate of outcomes classified as Continuation decreases monotonically with MFE\_pre. This pattern is statistically robust ( $Z=-7.697$ ,  $p<0.001$ ) and persists in the multivariate model in Section 8. However, as cautioned in Section 3.6, defining Continuation as exceeding MFE\_pre creates a mechanical relationship: the larger the MFE\_pre, the higher the bar to clear. We cannot determine what fraction of this pattern reflects a genuine market phenomenon and what fraction is a consequence of that mechanical relationship. We report the pattern because it is statistically consistent and because the question of what produces it is itself valuable, but the reader should keep this caveat in mind.

## 7.5 Secondary Associations

**Table 13 — Continuation Rate by Individual Predictors (OR = 15 minutes)**

| Variable    | Segment       | N   | %Cont | 95% CI         | %Failure |
|-------------|---------------|-----|-------|----------------|----------|
| OR_size_pct | Q1 minimum    | 209 | 29.2% | [23.5%, 35.7%] | 27.8%    |
|             | Q3 medium     | 209 | 49.8% | [43.2%, 56.4%] | 16.3%    |
|             | Q5 very large | 209 | 58.9% | [52.2%, 65.3%] | 14.4%    |
| Vol_ratio   | Q1 lowest     | 209 | 35.4% | [29.3%, 42.1%] | 17.2%    |
|             | Q3 optimal    | 209 | 56.9% | [50.2%, 63.3%] | 14.4%    |
|             | Q5 highest    | 207 | 46.9% | [40.3%, 53.7%] | 21.7%    |

OR\_size ( $\chi^2=55.26$ ,  $p<0.001$ ) and vol\_ratio ( $\chi^2=26.85$ ,  $p=0.00075$ ) show statistically significant associations with the outcome. For OR\_size, part of the pattern in the %Failure column may be a consequence of the mechanical definitional effect documented in Section 3.6. The %Cont column is not subject to that effect.

## 8. MULTIVARIATE ANALYSIS

### 8.1 Purpose

Levels 1 through 4 analyzed each variable independently. This analysis has two distinct objectives. The first is to verify whether the observed effects persist when mutually controlled — in particular, whether the association between mins\_to\_retest and the outcome is independent of breakout time, which is a potential source of confounding. The second is to measure the correlation among variables in order to assess whether they capture genuinely distinct dimensions of the phenomenon.

### 8.2 Logistic Models

We estimate four logistic models with the dependent variable equal to 1 for Continuation outcomes and 0 otherwise. All continuous variables were standardized — zero mean, unit standard deviation — so that coefficients are comparable across variables. The simplification of treating Failure and Neutral outcomes as a combined category is a methodological limitation: information about the distinction between the two is lost, which may bias the estimated probabilities.

**Table 14 — Logistic Model Comparison (OR = 15 minutes, N = 1,045)**

| <b>Model</b> | <b>Variables Included</b>        | <b>Pseudo-R<sup>2</sup></b> | <b>AIC</b> |
|--------------|----------------------------------|-----------------------------|------------|
| <b>M1</b>    | mins_to_retest only              | 0.0575                      | 1363.49    |
| <b>M2</b>    | mins_to_retest + breakout_minute | 0.0584                      | 1364.17    |
| <b>M3</b>    | Full model (5 variables)         | 0.1037                      | 1304.75    |
| <b>M4</b>    | Full model + quadratic terms     | 0.1071                      | 1305.94    |

M3 has the best AIC. The fact that adding breakout\_minute in M2 worsens the AIC relative to M1 indicates that this variable contributes no additional relevant information once mins\_to\_retest is in the model.

**Table 15 — Full Model M3 Coefficients (standardized variables)**

| <b>Variable</b>           | <b>Coef</b> | <b>SE</b> | <b>Z</b> | <b>p-value</b> | <b>Sig</b> | <b>OR</b> | <b>95% CI OR</b> |
|---------------------------|-------------|-----------|----------|----------------|------------|-----------|------------------|
| <b>mins_to_retest</b>     | −0.325      | 0.079     | −4.115   | <0.001         | ***        | 0.722     | [0.619, 0.843]   |
| <b>breakout_minute</b>    | −0.108      | 0.072     | −1.499   | 0.134          | n.s.       | 0.897     | [0.779, 1.034]   |
| <b>or_size_pct</b>        | +0.450      | 0.083     | +5.450   | <0.001         | ***        | 1.568     | [1.334, 1.844]   |
| <b>vol_ratio</b>          | +0.120      | 0.069     | +1.730   | 0.084          | n.s.       | 1.127     | [0.984, 1.291]   |
| <b>mfe_pre_retest_pct</b> | −0.622      | 0.099     | −6.311   | <0.001         | ***        | 0.537     | [0.442, 0.651]   |

Three variables show independent, statistically significant associations with the Continuation outcome: mins\_to\_retest (OR=0.722; each standard deviation increase in time to retest is associated with a 27.8% reduction in the odds of Continuation), or\_size\_pct (OR=1.568), and mfe\_pre\_retest\_pct (OR=0.537).

breakout\_minute is not significant in any model (p=0.257 in M2, p=0.134 in M3). This answers the critical question about potential confounding: the association between mins\_to\_retest and the outcome is not explained by the time of day at which the breakout occurred. The coefficient of mins\_to\_retest changes by less than 0.6% between M1 and M2 (from −0.612 to −0.608), confirming that breakout time was not biasing the estimate.

For `mfe_pre_retest_pct`, the logistic model confirms that the association persists when controlling for the other variables. However, as documented in Section 3.6, the mechanical relationship between `MFE_pre` and the Continuation definition would also be captured by this model. The statistical significance of the coefficient does not allow the genuine effect to be separated from the definitional effect.

No quadratic term is significant in M4 (all  $p > 0.07$ ), indicating that the relationships are approximately linear on the log-odds scale within the observed range of variation.

### 8.3 Correlations Among Variables

To assess whether the identified variables capture genuinely distinct dimensions, we computed the Phi correlation between their binarized versions.

**Table 16 — Phi Correlations Between Binarized Variables (OR = 15 minutes)**

| Pair                                  | $\phi$   | Level      |
|---------------------------------------|----------|------------|
| Large OR – Retest $\leq 10$ min       | 0.293    | Moderate   |
| Breakout $\leq 30$ min – Aligned bias | 0.245    | Moderate   |
| All other pairs (13 pairs)            | $< 0.10$ | Negligible |

The maximum observed correlation is  $\phi = 0.293$ . This value is below the conventional threshold of 0.30 for substantial correlation, suggesting that the variables operate relatively independently in the data.

However, the Phi correlation measures linear association between binarized variables. A low Phi correlation does not guarantee full conceptual independence: all of these variables derive from the same temporal process and are measured on the same price bars. They may share sources of variation that this analysis does not capture. This nuance is important to avoid over-interpreting the independence results.

## 9. ROBUSTNESS ANALYSIS

### 9.1 The Five Market Regimes

*Table 17 — Subperiod Definitions*

| Period    | Dates     | Regime                   | QQQ Return | N retests |
|-----------|-----------|--------------------------|------------|-----------|
| <b>P1</b> | 2017–2019 | Quiet bull market        | +122%      | 274       |
| <b>P2</b> | 2020      | COVID crash and recovery | +48%       | 151       |
| <b>P3</b> | 2021      | Accelerated bull market  | +27%       | 88        |
| <b>P4</b> | 2022      | Bear market              | −33%       | 170       |
| <b>P5</b> | 2023–2024 | Recovery to new highs    | +85%       | 233       |

### 9.2 Base Rates by Subperiod

*Table 18 — Base Metrics by Subperiod (OR = 15 minutes)*

| Subperiod              | N     | %Cont | 95% CI         | %Failure | %Neutral |
|------------------------|-------|-------|----------------|----------|----------|
| <b>TOTAL</b>           | 1,045 | 46.1% | [43.1%, 49.2%] | 18.7%    | 35.2%    |
| <b>P1 — 2017–2019</b>  | 274   | 39.8% | [34.3%, 45.6%] | 17.9%    | 42.3%    |
| <b>P2 — 2020 COVID</b> | 151   | 52.3% | [44.3%, 60.2%] | 16.6%    | 31.1%    |
| <b>P3 — 2021 bull</b>  | 88    | 37.5% | [27.9%, 48.2%] | 17.0%    | 45.5%    |
| <b>P4 — 2022 bear</b>  | 170   | 60.6% | [53.1%, 67.6%] | 19.4%    | 20.0%    |
| <b>P5 — 2023–2024</b>  | 233   | 42.5% | [36.5%, 48.8%] | 21.5%    | 36.1%    |

Subperiod P4 (2022) shows the highest Continuation rate (60.6%) and is statistically different from all other subperiods ( $p < 0.001$  in all tests). The three bullish or normal periods P1, P3, and P5 are statistically indistinguishable from one another. The mechanism behind P4's difference is unknown. We report the statistical difference as an observation without proposing a causal explanation.

### 9.3 Robustness of the Temporal Pattern Across Regimes

**Table 19 — Continuation Rate by mins\_to\_retest and Subperiod**

| Mins to Retest | TOTAL | P1    | P2    | P3    | P4    | P5    |
|----------------|-------|-------|-------|-------|-------|-------|
| 0–10 min       | 69.0% | 73.0% | 64.3% | 71.4% | 69.6% | 68.3% |
| 10–20 min      | 54.0% | 44.7% | 53.1% | 62.5% | 61.2% | 49.1% |
| 20–30 min      | 47.1% | 40.0% | 63.6% | 33.3% | 64.3% | 38.2% |
| 30–60 min      | 37.4% | 41.5% | 48.5% | 16.7% | 50.0% | 25.0% |
| >60 min        | 25.6% | 19.2% | 22.7% | 22.7% | 31.2% | 35.3% |

The 0–10 minute bin produces the highest Continuation rate in all five subperiods without exception, with a maximum variation of 8.7 percentage points across subperiods. The decreasing trend holds across all regimes studied.

### 9.4 Robustness of the MFE\_pre Pattern Across Regimes

**Table 20 — Continuation Rate by MFE\_pre and Subperiod**

| MFE_pre Bin | TOTAL | P1    | P2    | P3    | P4    | P5    |
|-------------|-------|-------|-------|-------|-------|-------|
| 0.20–0.30%  | 54.6% | 48.7% | 54.3% | 58.5% | 69.5% | 52.4% |
| 0.30–0.40%  | 46.3% | 38.1% | 64.3% | 26.1% | 71.1% | 38.1% |
| 0.40–0.50%  | 34.0% | 26.1% | 31.2% | N/A   | 60.9% | 30.4% |
| 0.50–0.75%  | 22.6% | 10.0% | 40.0% | 21.4% | 22.2% | 17.6% |

The decreasing trend is observed in four of the five subperiods. P2 (COVID 2020) is the most notable exception. The caveat regarding the mechanical definitional effect applies equally to each column of this table.

### 9.5 Robustness to Variations in Definition Thresholds

The correct metric for evaluating robustness to threshold variations is not the absolute Continuation rate — that varies because different thresholds include populations with different characteristics. The

correct metric is the relative gap between fast and slow retests, which compares the same relative pattern within each population.

**Table 21 — Robustness of the Fast vs. Slow Retest Gap Under Threshold Variations**

| <b>Varied Threshold</b>  | <b>Range Explored</b> | <b>Base Gap</b> | <b>Min Gap</b> | <b>Max Gap</b> | <b>Max Dev.</b> |
|--------------------------|-----------------------|-----------------|----------------|----------------|-----------------|
| <b>Minimum excursion</b> | 0.10% to 0.30%        | 41.8pp          | 39.5pp         | 45.5pp         | 3.7pp           |
| <b>Touch tolerance</b>   | 0.03% to 0.10%        | 41.8pp          | 36.7pp         | 41.3pp         | 5.1pp           |
| <b>Time window</b>       | 60 min to close       | 41.8pp          | 41.6pp         | 48.3pp         | 6.5pp           |

The gap ranges from 36.7 to 48.3 percentage points with a maximum deviation of 6.5pp. The Continuation rate associated with fast retests is exactly 68.1% across all four time window scenarios explored. These results suggest that the temporal pattern of the retest does not depend on the exact threshold values chosen.

## 9.6 Resolution of Neutral Cases

**Table 22 — Resolution of Neutral Cases Under Extended Windows (OR = 15 minutes, N = 368)**

| <b>Resolution</b>   | <b>Between 60 and 120 min</b> | <b>Through Session Close</b> |
|---------------------|-------------------------------|------------------------------|
| <b>Continuation</b> | 74 (20.1%)                    | 153 (41.6%)                  |
| <b>Failure</b>      | 77 (20.9%)                    | 146 (39.7%)                  |
| <b>Unresolved</b>   | 217 (59.0%)                   | 69 (18.8%)                   |

Neutral cases resolve approximately symmetrically between Continuation and Failure by the session close (41.6% versus 39.7%), under the same outcome definitions. This is consistent with the interpretation that Neutral cases represent genuine market indecision within the original 60-minute window, rather than continuations or failures that occurred later with a predominant direction.

# 10. DISCUSSION

## 10.1 On the Time-to-Retest Pattern

The pattern that faster retests are associated with higher Continuation rates is the most robust finding in this paper. It appears across all market regimes studied, survives logistic regression with multiple

controls, is independent of breakout time, and is not subject to the mechanical effect of the MFE\_pre definition.

One possible interpretive hypothesis, drawing on the microstructure concepts of Dim, Eraker, and Vilkov (2023), is that fast retests occur when participants have limit orders placed near the breakout level, waiting for a second entry opportunity. If those orders are sufficient to absorb the pullback pressure, price returns quickly to the level and resumes its direction. A late retest would occur in a context where that potential had been exhausted. This hypothesis is plausible as an interpretive framework, but was not verified directly in this paper. It remains a concrete direction for future research that would require order flow data not available in this study.

## **10.2 On the MFE\_pre Pattern**

The pattern that larger pre-retest excursions are associated with lower Continuation rates is statistically robust but methodologically more delicate than the temporal pattern. As documented with precision in Section 3.6, defining Continuation as exceeding MFE\_pre creates a mechanical relationship: the bar to clear grows with MFE\_pre. Any analysis using this definition will tend to show this pattern to some degree, regardless of what the market does.

What we cannot determine with the current design is what fraction of the observed pattern reflects a genuine market phenomenon and what fraction is a consequence of the definition. We report the pattern because it is consistent and because the question of its origin is itself valuable. The way to resolve it in future research would be to compare results using a Continuation definition based on a fixed target rather than a variable one. If the pattern persists under that alternative definition, it would constitute stronger evidence that it reflects a genuine market phenomenon.

## **10.3 On the Statistical Difference of the 2022 Subperiod**

Subperiod P4 (2022) shows a Continuation rate of 60.6%, statistically different from all other periods at the highest level of significance ( $p < 0.001$ ). We report this difference as a statistical observation without proposing a causal explanation. The mechanism is unknown and remains an open question for future research.

## **10.4 Relationship with the Existing Literature**

The Level 1 associations are consistent with Lundström (2017)'s result on ORB profitability scaling with volatility, extended to the level of the individual breakout. Level 2 is consistent with Tsai et al. (2018) on the concentration of price discovery at the open. The time-to-retest pattern has no direct precedent in the literature known to us, which represents the most novel contribution of this paper.

## 10.5 Limitations

*Limitation 1 — A single instrument over a single period.* The observed patterns are specific to QQQ over 2017–2024. Generalization to other instruments, markets, or periods requires independent replication. This paper provides no basis for assuming that the same patterns, with the same numerical values, would appear in another instrument or period.

*Limitation 2 — No transaction costs.* The analysis is conducted on the underlying price. No result should be interpreted as evidence of profitability in live trading. Bid-ask spreads, execution slippage, and commission costs are not incorporated.

*Limitation 3 — The effect of definitions on results.* As documented in detail in Section 3.6, two of the observed patterns may be influenced by the chosen definitions. Continuation defined as exceeding MFE\_pre mechanically tends to produce lower rates when MFE\_pre is large. Failure defined as crossing the opposite OR extreme mechanically tends to produce lower rates when the range is wide. We cannot precisely quantify how much of the numerical patterns observed for those variables is attributable to these effects.

*Limitation 4 — Multiple testing.* The analysis evaluated multiple variables, bins, OR durations, and subperiods without correction for multiple comparisons. Some results may be statistically significant by chance. The most reliable patterns are those that appear consistently across multiple specifications.

*Limitation 5 — Selection bias.* Only the first breakout of the day is analyzed, and only retests that satisfy the three defined conditions. The analyzed sample does not represent all QQQ trading days but a filtered subset. The documented patterns apply to that specific subset.

*Limitation 6 — Heterogeneous sample sizes across subperiods.* P3 (2021) contains only 88 retests with a 15-minute OR. Estimates for that subperiod have wide confidence intervals and should be treated as indicative.

*Limitation 7 — Temporal independence not formally verified.* Daily observations may exhibit serial autocorrelation. Autocorrelation-robust standard errors were not applied, which may artificially inflate the reported statistical significance.

*Limitation 8 — Mechanisms are unverified hypotheses.* All causal interpretations proposed in Section 10 are plausible hypotheses grounded in microstructure literature. They were not verified using order flow data or any other method independent of the one-minute prices used in this paper.

## 11. CONCLUSION

This paper documents, in an exploratory and descriptive manner and under explicit operational definitions, four statistical associations in the post-breakout sequence of the ORB in QQQ, using intraday data from 2017 to 2024.

We make no claims of causality. We make no claims of exploitability. We make no claims that these patterns will persist outside the sample or replicate in other instruments. We do claim that we measured them with the greatest care available to us, that we documented the ways in which our own definitions may have influenced the results, and that we present this work as a first step in a direction we consider valuable for research.

The four documented associations are as follows.

First, opening range size shows a linear and monotonic association with the observed maximum post-breakout excursion, with a 2.8-fold difference between the minimum and maximum quintiles ( $Z=+7.109$ ,  $p<0.001$ ). This pattern holds without exception across all 24 range durations studied.

Second, under the retest conditions defined in this paper, 68.4% of breakouts generate a retest of the broken level within 120 minutes. The probability of that retest occurring grows monotonically with the price excursion prior to the retest and approaches 100% when that excursion exceeds 0.50%.

Third, the time elapsed between the breakout and the retest shows the strongest and most robust association with the outcome classified as Continuation. Retests within 10 minutes are associated with a rate of 69.0% [95% CI: 62.5%, 74.8%] versus 25.6% [20.3%, 31.6%] for retests after 60 minutes ( $\chi^2=170.67$ ,  $p<0.001$ ). Logistic regression confirms that this association is independent of breakout time (OR=0.722,  $p<0.001$ ; breakout\_minute not significant in any model). The pattern holds across all five market regimes studied.

Fourth, larger pre-retest excursions are associated with lower Continuation outcome rates ( $Z=-7.697$ ,  $p<0.001$ ; OR=0.537 in the multivariate model). This pattern is statistically robust but must be interpreted with particular caution, as the Continuation definition used may mechanically produce part of this result regardless of market behavior.

We present this work as an invitation to continue. The most important questions we leave open are the following. Do these patterns replicate in SPY, SPX, IWM, and other liquid instruments, or are they specific to QQQ in this period? Does the MFE\_pre pattern persist when Continuation is redefined using a fixed rather than variable target? What explains the statistical difference of the 2022

subperiod? Do data available from 2025 onward — not included in this analysis — confirm or contradict the documented associations?

We hope that other researchers find in this paper a useful starting point, not a conclusion.

Data, code, and complete results tables are available from the authors upon request.

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